LAB 5 and 6

Objectives:

1. **WAP to find the greatest number among tow 8-bit number.**
2. **WAP to find the smallest number among two 8-bit number.**
1. **WAP to find whether a number is odd or even.**
2. **WAP to display all odd numbers from 1 to 10.**
3. **WAP to display all even numbers from 1 to 10.**
4. **WAP to display number from 1 to 10.**
5. **WAP to find the sum of numbers from 1 to 10 and store the result in memory location C055.**

Program to find greatest among two 8-bit numbers.

Statement: Input numbers from memory location C050H & C051H and store greatest number in memory location C055H.

Address	Hex Code (Opcode)	Mnemonics	Comments
C000	3A	LDA C051	Load contents from memory location C051H to accumulator
C001	51		
C002	C0		
C003	47	MOV B,A	Move contents of Accumulator to register B
C004	3A	LDA C050	
C005	50		
C006	C0		
C007	B8	CMP B	The contents of the Register B is compared with the contents of the accumulator and both contents are preserved
C008	D2	JNC X	Jump if no carry to X
C009	0C		
C00A	C0		
C00B	78	MOV A,B	Move contents of register B to Accumulator
C00C	32	X: STA CO55	Store result in memory location C055H
C00D	55		
C00E	C0		
C00F	76	HLT	Terminate the program.

Input:

Memory Location Data

C050: 10H

C051: 20H

Output:

Memory Location Data

C055: 20H

Program to find smallest among two 8-bit numbers.

Statement: Input numbers from memory location C050H & C051H and store smallest number in memory location C055H.

Address	Hex Code (Opcode)	Mnemonics	Comments
C000	3A	LDA C051	Load contents from memory location C051H to accumulator
C001	51		
C002	C0		
C003	47	MOV B,A	Move contents of Accumulator to register B
C004	3A	LDA C050	Load contents from memory location C050H to accumulator
C005	50		
C006	C0		
C007	B8	CMP B	The contents of the Register B is compared with the contents of the accumulator and both contents are preserved
C008	D2	JC X	Jump if carry to X
C009	0C		
C00A	C0		
C00B	78	MOV A,B	Move contents of register B to Accumulator
C00C	32	X: STA CO55	Store result in memory location C055H
C00D	55		
C00E	C0		
C00F	76	HLT	Terminate the program.

Input:

Memory Location Data

C050: 10H

C051: 20H

Output:

Memory Location Data

C055: 10H

Program to find whether a number is odd or even.

Statement: Input number from memory location C050H and store result in C055H.

Address	Hex Code (Opcode)	Mnemonics	Comments
C000	3A	LDA C050	Load contents from memory location C050H to accumulator
C001	50		
C002	C0		
C003	E6	ANI 01	The contents of the accumulator are logically ANDed with the 01H. CY is reset, AC is set.
C004	01		
C005	CA	JZ X	Jump if zero to label X
C006	0D		
C007	C0		
C008	3E	MVI A, 0D	Loads the accumulator with 0D
C009	0D		
C00A	C3	JMP Y	Jump Unconditionally
C00B	0F		
C00C	C0		
C00D	3E	X: MVI A, 0E	Loads the accumulator with 0E
C00E	0E		
C00F	32	Y: STA C055	Store result in memory location C055H
C010	55		
C011	C0		
C012	76	HLT	Terminate the program.

Input:

Memory Location Data

C050: 10H

Output:

Memory Location Data

C055: 0EH

Program to display all odd numbers from 1 to 10.

Address	Hex Code (Opcode)	Mnemonics	Comments
C000	21	LXI H, C050	Load 16-bit data C050H to HL pair
C001	50		
C002	C0		
C003	06	MVI B, 01	Loads the register B with immediate data 01 H
C004	01		
C005	0E	MVI C, 0A	Loads the register C with immediate data 0A H
C006	0A		
C007	70	X: MOV M,B	Move contents of register B to M

C008	23	INX H	The contents of register pair H-L is incremented by 1
C009	04	INR B	Increment the register B by 1
C00A	04	INR B	Increment the register B by 1
C00B	0D	DCR C	Decrement the register C by 1
C00C	0D	DCR C	Decrement the register C by 1
C00D	C2	JNZ X	Jump if no zero to X
C00E	07		
C00F	C0		
C010	76	HLT	Terminate the program.

Output:

Memory Location Data

C050 = 01H

C051 = 03H

C052 = 05H

C053 = 07H

C054 = 09H

Write a program to display all even numbers from 1 to 10.

Address	Hex Code (Opcode)	Mnemonics	Comments
C000	21	LXI H, C050	Load 16-bit data C050H to HL pair
C001	50		
C002	C0		
C003	06	MVI B, 02	Loads the register B with immediate data 02 H
C004	02		
C005	0E	MVI C 0A	Loads the register C with immediate data 0A H
C006	0A		
C007	70	X: MOV M, B	Move contents of register B to M
C008	23	INX H	Increment the register pair H-L by 1
C009	04	INR B	Increment the register B by 1
C00A	04	INR B	Increment the register B by 1
C00B	0D	DCR C	Decrement the register C by 1
C00C	0D	DCR C	Decrement the register C by 1
C00D	C2	JNZ X	Jump if no zero to X
C00E	07		
C00F	C0		
C010	76	HLT	Terminate the program.

Output:

Memory Location Data

C050 = 02H

C051 = 04H

C052 = 06H

C053 = 08H

C054 = 0AH

Program to display number from 1 to 10.

Address	Hex Code (Opcode)	Mnemonics	Comments
C000	21	LXI H, C050	Load 16-bit data C050H to HL pair
C001	50		
C002	C0		
C003	06	MVI B, 01	Loads the register B with immediate data 01 H
C004	01		
C005	0E	MVI C, 0A	Loads the register C with immediate data 0A H
C006	0A		
C007	70	X: MOV M, B	Move contents of register B to M
C008	23	INX H	The contents of register pair H-L are incremented by 1
C009	04	INR B	Increment the content of register B by 1
C00A	0D	DCR C	Decrement the content of register C by 1
C00B	C2	JNZ X	Jump if no zero to X
C00C	07		
C00D	C0		
C00E	76	HLT	Terminate the program.

Output:

Memory Location Data

C050 = 01H

C051 = 02H

C052 = 03H

C053 = 04H

C054 = 05H

C055 = 06H

C056 = 07H

C057 = 08H

C058 = 09H

C059 =0AH

Program to find the sum of numbers from 1 to 10 and store the result in memory location C055.

Address	Hex Code (Opcode)	Mnemonics	Comments
C000	21	LXI H, C050	Load 16-bit data C050H to HL pair
C001	50		
C002	C0		
C003	06	MVI B, 01	Loads the register B with immediate data 01 H
C004	01		
C005	0E	MVI C, 0A	Loads the register C with immediate data 0A H
C006	0A		
C007	3E	MVI A, 00	Loads the accumulator with immediate data 00 H
C008	00		
C009	80	X: ADD B	Add the content of register B with accumulator and store the result in accumulator
C00A	23	INX H	The contents of register pair H-L is incremented by 1
C00B	04	INR B	Increment the content of register B by 1
C00C	0D	DCR C	Decrement the content of register C by 1
C00D	C2	JNZ X	Jump if no zero to X
C00E	09		
C00F	C0		
C010	32	STA C055	Store result in memory location C055H
C011	55		
C012	C0		
C013	76	HLT	Terminate the program.

Output:

Memory Location Data

C055 = 37H